

Student Assignment Submission

- **Student Name:** Osama Ahmed Sayed
 - **Assignment Title:** Freelance Project Plan and Delivery Roadmap
 - **Session:** Session 3 - Essentials of Project Management & Project Management Planning and Scope
 - **Total Marks:** 20
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1. Project Overview & Requirements

- **Project Title:** E-Commerce RESTful API Backend Development
- **Client Requirement:** The client runs a growing retail business and requires a robust, scalable backend RESTful API built with ASP.NET Core and SQL Server to manage user authentication, product catalogs, shopping cart operations, and secure order processing. The system must replace their legacy manual tracking sheets with an automated, high-performance web service.
- **Project Objective:** Design, develop, test, and deploy a secure, well-documented backend API that handles up to 500 concurrent users with sub-500ms response times, ensuring secure data persistence and seamless frontend integration.
- **Assumptions:**
 - The client will provide the relational database schema and business rule documentation within 2 days of project kickoff.
 - Third-party payment gateway sandbox credentials (e.g., Paymob/Stripe) will be provided promptly for integration testing.
 - Hosting infrastructure (such as Railway or MonsterASP.NET) will be configured and made accessible for staging deployment.

2. Project Scope

- **In-Scope:**
 - User authentication and authorization (JWT-based secure login, registration, and role management for Admin and Customers).
 - Product management endpoints (CRUD operations for categories, items, inventory levels, and search filters).
 - Cart and order management system (adding items, applying discounts, order placement, and status updates).
 - SQL Server database design, migration scripts, and Entity Framework Core integration.
 - Comprehensive API documentation via Swagger/OpenAPI.
- **Out-of-Scope:**
 - Development of the customer-facing frontend user interface (UI/UX design and React/Angular frontend implementation).
 - Native mobile application development (iOS/Android apps).

- Advanced AI-driven product recommendation engines.

3. Measurable Deliverables

1. Fully functional, secure ASP.NET Core Web API source code repository hosted on GitHub.
2. Executable SQL Server database migration scripts and initial seed data.
3. Interactive API documentation via Swagger UI covering all endpoints and response codes.
4. Postman collection containing automated test scripts for all primary API workflows.
5. Deployment guide and environment configuration documentation.

4. Work Breakdown Structure (Tasks, Milestones & Timeline)

Milestone	Task Description	Dependencies	Deadline
Milestone 1: Architecture & Setup	Database schema design, repository structure setup, and EF Core configuration.	Kickoff & Requirements Sign-off	Day 3
Milestone 2: Core API Development	Implement Authentication (JWT) and Product/Inventory CRUD endpoints.	Milestone 1	Day 7
Milestone 3: Advanced Features	Implement Shopping Cart, Order Processing workflows, and error handling middleware.	Milestone 2	Day 11
Milestone 4: Testing & Documentation	Write unit tests, run integration checks, finalize Postman collection, and configure Swagger.	Milestone 3	Day 13
Milestone 5: Deployment & Handoff	Deploy to staging environment, conduct final client walkthrough, and transfer repository ownership.	Milestone 4	Day 14

5. Risk Management & Mitigation

- **Key Project Risk:** Delays in receiving third-party payment gateway sandbox credentials or changes in business logic requirements mid-development.
- **Practical Mitigation Action:** Establish a modular architecture where the core inventory and order placement logic are decoupled from the payment gateway layer. If credentials are delayed, mock the payment response service to maintain development momentum, allowing payment integration to be plugged in seamlessly once credentials are provided.

6. Client Progress-Update Message

Subject: Weekly Project Status Update: E-Commerce API Backend Development

Dear [Client Name],

I hope this message finds you well. I am writing to provide you with a brief update on the progress of your E-Commerce API backend project for this week.

Progress Made:

- * Completed the database schema design and Entity Framework Core migrations.
- * Successfully implemented the JWT authentication module and core product catalog CRUD endpoints.

Next Steps:

- * We are moving into Milestone 3 to build out the shopping cart and secure order processing workflows.

Action Required:

- * Kindly review the attached preliminary Swagger documentation link to confirm that the product filter parameters align with your expectations.

Please let me know if you have any questions or feedback.

Best regards,
Osama Ahmed Sayed
Backend Software Developer

7. Quality, Final-Delivery, Revisions, and Project-Closure Checklist

- **Review Phase:** Conduct internal code reviews, check coding conventions, and verify

adherence to C# and ASP.NET Core best practices.

- **Testing Phase:** Run automated unit tests and Postman integration scripts to ensure zero critical bugs across all endpoints.
- **Handoff & Deployment:** Deploy the stable release to the staging server and share access credentials and documentation with the client.
- **Revisions Management:** Address one round of client feedback and incorporate requested minor adjustments within 48 hours.
- **Client Sign-off:** Obtain formal written approval and sign-off on deliverables from the client.
- **Project Closure:** Transfer final GitHub repository ownership, archive local staging instances, and provide final project sign-off documentation.